

CELL AND MOLECULAR BIOLOGY LABORATORY

From Gene Mutation to Disease:

Investigating How DNA Sequence Changes Affect Protein Products and Human Phenotypes

Student: Deguit, Zenas Charis C.	Documented Variant: BCKDHA c.929C>G (p.Thr310Arg / T310R)
Disease: Maple Syrup Urine Disease (MSUD)	Instructors: Ma'am Lilibeth Bucol & Sir Abner Bucol
Gene: BCKDHA	Date: September 16, 2026

I. INTRODUCTION

1.1 Background

Maple Syrup Urine Disease (MSUD) is an inherited metabolic disorder caused by impaired breakdown of the branched-chain amino acids leucine, isoleucine, and valine. The disorder occurs when the branched-chain α -ketoacid dehydrogenase (BCKD) complex does not function properly. As a result, branched-chain amino acids and their corresponding metabolites can accumulate in the body, particularly affecting the nervous system.

MSUD is commonly associated with symptoms such as poor feeding, vomiting, lethargy, abnormal muscle tone, neurological problems, seizures, and developmental difficulties. In severe cases, untreated metabolic abnormalities can progress to encephalopathy and coma. The disease can vary in severity depending on the extent of the remaining BCKD enzyme activity.

The BCKDHA gene is one of the genes involved in MSUD. It encodes the E1 α subunit of the BCKD complex, which is required for the normal metabolism of branched-chain amino acids. Mutations in BCKDHA can alter the structure or function of this subunit and consequently reduce the activity of the BCKD complex. The resulting disruption in amino-acid metabolism contributes to the biochemical abnormalities associated with MSUD.

MSUD is inherited in an autosomal recessive manner, meaning that an affected individual generally inherits a disease-causing variant from both parents. Understanding the relationship between mutations in genes such as BCKDHA, changes in the encoded protein, and the resulting metabolic effects is important for

explaining how a change at the DNA level can contribute to a human disease phenotype.

II. DISEASE AND GENE BACKGROUND

A. Disease

13. What is the disease or phenotype?

- Maple Syrup Urine Disease (MSUD) is an inherited metabolic disorder in which the body can't properly break down the branched-chain amino acids which are the leucine, isoleucine, and valine. Their breakdown products can accumulate in the body, particularly affecting the brain and nervous system.

14. What are its major clinical characteristics?

- The major clinical characteristics of Maple Syrup Urine Disease (MSUD) can include: Sweet or maple-syrup-like odor that can be detected in the urine and other body fluids due to the accumulation of branched-chain amino acids and their metabolites; Poor feeding, especially in affected infants, making it difficult for them to consume enough food; Vomiting; Irritability, result of metabolic disturbances affecting the nervous system; Lethargy, or unusual tiredness and reduced activity; Abnormal movements or muscle tone because of the accumulation of toxic metabolites that can interfere with normal neurological function; Developmental delay; Encephalopathy, or dysfunction of the brain, occur during periods of metabolic decompensation; Seizures, occur when the nervous system is severely affected. In severe, untreated cases, MSUD can progress to coma, which reflects serious neurological and metabolic dysfunction. The severity of MSUD varies according to its clinical form. Classic MSUD usually presents during the neonatal period and is generally the most severe form. Intermediate and intermittent forms may appear later in childhood or adulthood and can produce less severe or episodic symptoms.

15. Which cells, tissues, or organs are mainly affected?

- The nervous system, particularly in the brain, is an important target of the disease because accumulation of branched-chain amino acids and their metabolites can be toxic to cells and tissues. Severe metabolic intoxication can result in encephalopathy and other neurological problems.

16. What is its genetic basis?

- Maple Syrup Urine Disease (MSUD) is caused primarily by pathogenic variants in genes encoding components of the branched-chain α -ketoacid dehydrogenase (BCKD) α -ketoacid

dehydrogenase (BCKD) complex. The major genes associated with MSUD are BCKDHA, BCKDHB, and DBT. When its function is reduced or lost, these amino acids and their metabolites accumulate.

- **17. What is its inheritance pattern, if applicable?** Maple Syrup Urine Disease (MSUD) is inherited in an autosomal recessive pattern. An affected individual generally has pathogenic variants in both copies of the relevant gene. Parents of an affected individual are typically carriers of one pathogenic variant and do not show the disease.

B. Gene and Normal Protein

18. What is the official gene symbol?

- The official gene symbol is BCKDHA, which stands for branched chain keto acid dehydrogenase E1 subunit alpha.

19. On which human chromosome is the gene located?

- The BCKDHA gene is located on chromosome 19, specifically at 19q13.2

20. What does the gene normally encode?

- The BCKDHA gene encodes the alpha subunit of the E1 component of the branched-chain alpha-keto acid dehydrogenase (BCKD) complex. This enzyme complex is involved in the breakdown of the branched-chain amino acids leucine, isoleucine, and valine.

21. What is the normal biological function of the protein?

- The BCKDHA protein forms part of the BCKD enzyme complex, which catalyzes the second major step in the breakdown of the branched-chain amino acids leucine, isoleucine, and valine. The alpha subunit is part of the E1 component of this enzyme complex and is required for normal branched-chain amino-acid metabolism.

22. Where in the cell is the protein normally found?

- The BCKDHA protein is a component of the mitochondrial BCKD complex. Therefore, it is associated

with the mitochondria, where the BCKD complex carries out branched-chain amino-acid catabolism.

23. In what biological pathway or cellular process does it participate?

- BCKDHA participates in the catabolism of branched-chain amino acids (BCAAs), specifically the breakdown of leucine, isoleucine, and valine. It functions as part of the mitochondrial BCKD complex, which is involved in the metabolism of these amino acids.

C. Documented Mutation

The BCKDHA c.929C>G (p.Thr310Arg) variant is a pathogenic mutation associated with Maple Syrup Urine Disease (MSUD). It involves a single change in the DNA sequence, where cytosine (C) at coding position 929 is replaced by guanine (G). This changes the original ACA codon to AGA, resulting in a change from threonine (T) to arginine (R) at amino-acid position 310. Because the mutation causes one amino acid to be replaced by another, it is classified as a missense variant (NCBI ClinVar, 2026).

Table 1. Documented BCKDHA c.929C>G (p.Thr310Arg) Variant Information

Variant Information	Details
Gene	BCKDHA
Reference transcript	NM_000709.4
Reference protein	NP_000700.1
Exact variant notation	NM_000709.4:c.929C>G (p.Thr310Arg)
Nucleotide change	C → G at c.929
Codon change	ACA → AGA

Predicted protein change	p.Thr310Arg (T310R)
Mutation type	Missense

Nucleotides affected	1 bp
ClinVar Variation ID	2381
ClinVar accession	VCV000002381.13
Clinical interpretation	Pathogenic
Associated condition	Maple Syrup Urine Disease (MSUD)

Studies have found that the T310R substitution can affect the normal function of the BCKD E1-alpha protein. The change may interfere with protein stability, folding, and interactions between the enzyme's subunits. This can greatly reduce BCKD enzyme activity and affect the breakdown of branched-chain amino acids (Wynn et al., 1998; NCBI ClinVar, 2026).

III. NORMAL REFERENCE SEQUENCE

For the wild-type reference of BCKDHA, the NCBI RefSeq database was used to obtain the corresponding coding sequence. The transcript selected for the analysis was NM_000709.4, which encodes the mitochondrial BCKD E1-alpha protein represented by NP_000700.1. Since the purpose of the next analysis is to translate the DNA sequence into its corresponding amino-acid sequence, only the coding region (CDS) was used rather than the entire transcript. The downloaded nucleotide sequence was kept unchanged to serve as the WT reference throughout the mutation analysis.

Table 2. Reference Sequence Information for the BCKDHA Gene

Reference Information	Details
Gene	BCKDHA
Reference transcript	NM_000709.4
Reference CDS	NM_000709.4 CDS
Reference protein	NP_000700.1
Protein	2-oxoisovalerate dehydrogenase subunit alpha, mitochondrial isoform 1 precursor

Reference Information	Details
Gene	BCKDHA
Sequence source	NCBI RefSeq
Sequence type used for translation	Nucleotide CDS

The downloaded BCKDHA nucleotide CDS was saved in FASTA format and preserved as the original WT sequence. This sequence will later be uploaded to Galaxy for translation and will also provide the reference sequence for introducing and examining the documented c.929C>G (p.Thr310Arg) mutation. The FASTA record identifies the sequence as the BCKDHA CDS and links it to NM_000709.4 and NP_000700.1.

IV. WILD-TYPE SEQUENCE ANALYSIS

The wild-type (WT) sequence used in this analysis corresponds to the normal coding sequence of the BCKDHA gene. The sequence was obtained from the NCBI RefSeq record NM_000709.4 and contains the protein-coding region used for translation. It is associated with the reference protein NP_000700.1. The WT sequence was kept unchanged so that it could serve as the baseline for comparison with the mutant sequence later in the analysis.

Parameter	Wild-Type BCKDHA Sequence
Gene	BCKDHA
Reference transcript	NM_000709.4
Reference protein	NP_000700.1
Sequence type	Coding DNA sequence (CDS)
Sequence format	FASTA
Organism	<i>Homo sapiens</i>
Sequence used	Normal/WT BCKDHA CDS

Mutation to be introduced later	c.929C>G
Expected protein change	p.Thr310Arg (T310R)

V. WILD-TYPE TRANSLATION

The normal BCKDHA coding sequence (CDS) was used as the wild-type (WT) control. The WT CDS was translated into its predicted amino-acid sequence using the SeqKit Translate tool in Galaxy with the standard genetic code and reading frame 1. The resulting protein sequence was saved as MSUD_protein.fasta

Table 3. *Wild-Type BCKDHA Sequence and Translation Results*

WT Sequence Information	Result
Gene	BCKDHA
Reference transcript/CDS	NM_000709.4
Reference protein	NP_000700.1
CDS length	1,338 bp
Predicted protein length	445 amino acids
Start codon	ATG
Stop codon	TGA
Reading frame	Frame 1
First 10 amino acids	MAVAIAAARV
Last 10 amino acids	EHYPLDHFDK
Protein FASTA filename	MSUD_protein.fasta

The WT BCKDHA CDS produced a predicted protein sequence of 445 amino acids. The CDS is 1,338 bp long, corresponding to 446 codons when the terminal stop codon is

included. Translation begins with the start codon ATG and terminates with the stop codon TGA. The predicted protein sequence was then used as the WT reference for the subsequent documented mutation experiment.

VI. Mutation Hypothesis

Table 4. *Predicted Effects of the BCKDHA c.929C>G Mutation*

Mutation Prediction	Expected Result
Mutation	BCKDHA c.929C>G (p.Thr310Arg / T310R)
Exact nucleotide change	C → G at coding position 929
Number of nucleotides affected	1 nucleotide (1 bp)
Predicted mutation type	Missense substitution / single-nucleotide variant
Predicted codon change	ACA → AGA
Predicted amino-acid change	Threonine (T) → Arginine (R) at position 310
Predicted effect on reading frame	No change; reading frame remains intact
Predicted effect on protein length	Expected to impair BCKDHA function and reduce the activity of the branched-chain α -keto acid dehydrogenase complex

Hypothesis: Introducing the documented BCKDHA c.929C>G variant into the WT CDS is expected to produce a Thr310Arg (T310R) amino-acid substitution while maintaining the original reading frame and protein length. Since only one nucleotide is substituted, the mutation is not expected to cause a frameshift or premature stop codon; instead, the main predicted effect is a change in the amino acid at position 310.

VII. DOCUMENTED MUTATION EXPERIMENT

Table 5. Documented BCKDHA c.929C>G Mutation Construction

Required Information	Result
Original nucleotide position	929

Original sequence	C
Mutant sequence	G
Original codon	ACA
Mutant codon	AGA
Bases inserted	0
Bases deleted	0
Bases substituted	1
Mutation type	Substitution (missense)

VIII. MUTANT SEQUENCE TRANSLATION

Table 6. Mutant BCKDHA Sequence Translation Results

Required Information	Result
Mutant CDS length	1,338 bp
Mutant protein length	445 amino acids
Reading frame	Frame 1
First amino-acid difference	Amino acid 310 (T → R)
Premature stop codon	Absent
Approximate number of amino acids affected	1 amino acid

Mutant protein FASTA filename MSUD_mutant_protein.fasta

The mutant BCKDHA CDS remained 1,338 bp long and produced a predicted protein of 445 amino acids, which was the same length as the WT protein. The first sequence difference occurred at amino-acid position 310, where threonine (T) was replaced by arginine (R). No reading-frame change or premature stop codon was observed, consistent with a single-nucleotide missense substitution.

IX. WT AND MUTANT PROTEIN COMPARISON

24. At what amino-acid position do the sequences first differ?

- The first difference between the WT and mutant sequences occurs at amino-acid position 310, where threonine (T) is replaced by arginine (R).

25. Is only one amino acid affected?

- Yes. The change is limited to one amino acid, resulting in the T310R substitution.

26. Are multiple downstream amino acids changed?

- No. The amino-acid sequence following position 310 remains unchanged in the mutant protein.

27. Was an amino acid deleted or inserted?

- No. The mutation does not involve the removal or addition of any amino acids.

28. Was a premature stop codon produced?

- No. The mutation did not introduce a premature termination codon, so translation continues to the normal stop codon.

29. Did the reading frame change?

- No. The reading frame is maintained because the mutation involves the replacement of only one nucleotide rather than an insertion or deletion.

30. Did the protein length change?

- No. Both the WT and mutant predicted proteins have a length of 445 amino acids.

31. Is the mutation missense, nonsense, frameshift, in-frame deletion/insertion, repeat expansion, or another type?

- The variant is classified as a missense mutation, specifically BCKDHA c.929C>G (p.Thr310Arg), because a single nucleotide change results in the substitution of one amino acid.

X. MOLECULAR CONSEQUENCE OF Q188R

The documented BCKDHA c.929C>G variant involves a change of one nucleotide in the coding region. In the WT sequence, position 929 contains C, while the mutant sequence has G at the same position. This changes the codon from ACA to AGA, which results in the replacement of threonine (T) with arginine (R) at amino-acid position 310. The SeqKit translation confirmed that the mutation does not alter the reading frame or protein length, since both the WT and mutant sequences produce a predicted protein of 445 amino acids and no premature stop codon is introduced.

Although the length of the protein remains the same, the amino-acid substitution can still affect the properties of the protein. Threonine and arginine have different chemical characteristics, so replacing one with the other may influence the protein's structure or stability. Wynn et al. (1998) reported experimental evidence for the corresponding T265R variant, an earlier numbering designation for the same mutation, showing that the altered E1 α subunit had reduced stability and impaired assembly with the E1 β subunit. The study also reported no detectable enzyme activity for the mutant complex, supporting the possibility that the amino-acid change interferes with the formation and function of the BCKD enzyme complex. This variant is also documented in NCBI ClinVar as NM_000709.4(BCKDHA):c.929C>G (p.Thr310Arg).

The BCKDHA gene encodes the E1 α subunit of the branched-chain α -ketoacid dehydrogenase (BCKD) complex. This complex is involved in the breakdown of the branched-chain amino acids leucine, isoleucine, and valine. If BCKD activity is reduced, these amino acids and their related α -ketoacids are not metabolized efficiently and can accumulate in the body. GeneReviews describes this accumulation as a major biochemical consequence of impaired BCKD activity in Maple Syrup Urine Disease (MSUD).

Overall, the sequence analysis shows the progression from the c.929C>G nucleotide substitution, to the ACA→AGA codon change, and then to the Thr310Arg amino-acid substitution. Experimental evidence suggests that this protein change can interfere with BCKDHA stability and assembly, which can reduce the activity of the BCKD complex. Reduced BCKD activity can then lead to the accumulation of branched-chain amino acids and their metabolites, contributing to the metabolic and neurological manifestations associated with MSUD. Thus, the computational results identify the specific DNA and protein changes, while the published experimental evidence helps explain how the amino-acid substitution can affect protein function and contribute to the disease phenotype.

XI. ARTIFICIAL MUTATION EXPERIMENT

The artificial mutation was created by making a single nucleotide change in the WT BCKDHA CDS at c.303 (G→A). This change converted the original AAG codon into AAA. Based on the genetic code, both codons specify lysine (K), so the mutation was expected to have no effect on the amino-acid sequence. As a result, the protein was predicted to maintain the same reading frame and overall length, with no amino-acid substitution or premature termination.

Table 7. *Artificial BCKDHA c.303G>A Mutation Information*

Artificial Mutation Information	Result
Gene	BCKDHA
Reference transcript	NM_000709.4
Mutation type	Single-nucleotide substitution
Nucleotide position	c.303
Nucleotide change	G → A
Codon change	AAG → AAA

Predicted amino acid change	K → K
Predicted mutation type	Synonymous
Predicted frameshift	No
Predicted premature stop	No
Predicted protein length	445 amino acids

XII. COMPARISON OF DOCUMENTED AND ARTIFICIAL MUTATIONS

Table 8. Comparison of Wild-Type, Documented Mutant, and Artificial Mutant BCKDHA Sequences

Feature	WT	Documented Mutation	Student-Created Mutation
CDS length	1,338 bp	1,338 bp	1,338 bp
Protein length	445 aa	445 aa	445 aa
Mutation	None	c.929C>G	c.303G>A
Codon change	ACA	ACA → AGA	AAG → AAA
Mutation type	N/A	Missense	Synonymous
Reading frame changed?	No	No	No
Premature stop codon?	No	No	No
Amino acids affected	None	T310 → R310	None (K → K)
Expected functional consequence	Normal BCKDHA protein function	May affect BCKDHA stability/assembly and reduce BCKD complex activity	No change expected in the predicted protein sequence; functional effects were not tested

The documented and artificial mutations produced different effects on the predicted BCKDHA protein even though both involved a single-nucleotide substitution. The documented **c.929C>G** mutation changed the codon from **ACA to AGA**, resulting in a **threonine-to-arginine substitution at amino acid 310 (T310R)**. In comparison, the artificial **c.303G>A** mutation changed **AAG to AAA**, but both codons encode lysine. Therefore, the artificial mutation was **synonymous** and did not alter the predicted amino-acid sequence. Both mutations maintained the same CDS length and protein length and did not cause a frameshift or premature stop codon.

XIII. Interpretation Questions

32. Why does the exact location of a mutation matter?

- The position of a mutation determines which part of the genetic sequence is affected. Depending on its location, the mutation may change a specific amino acid or interfere with an important region of the protein. Therefore, mutations in functionally important areas can have a greater effect on protein activity.

33. Why can deleting three nucleotides produce a different result from deleting one or two nucleotides?

- A deletion of three nucleotides removes one complete codon, so the remaining sequence can still be translated in the original reading frame. However, deleting one or two nucleotides changes the grouping of the following nucleotides into codons, causing a frameshift that can alter many downstream amino acids.

34. Does every mutation change the amino-acid sequence? Explain.

- No. Some nucleotide changes do not result in an amino-acid change because different codons can code for the same amino acid. In our artificial **BCKDHA** mutation, **AAG was changed to AAA**, but both codons specify lysine (K). Therefore, the predicted protein sequence remained unchanged.

35. Does every amino-acid substitution destroy protein function? Explain.

- No. An amino-acid substitution does not automatically make a protein nonfunctional. Its effect depends on the type of amino-acid change and where it occurs in the protein. Some substitutions may have minimal effects, whereas others can interfere with protein folding, stability, or activity.

36. Why can a frameshift affect many amino acids even if only one nucleotide was deleted?

- Removing one nucleotide shifts the reading frame of the sequence beginning at the mutation site. Because the nucleotides are then grouped into different sets of three, the codons after the deletion can produce a different series of amino acids. This can continue until a stop codon is encountered.

37. Why might a premature stop codon produce a nonfunctional protein?

- A premature stop codon causes translation to end earlier than it should. This produces a shortened protein that may not contain all of the regions necessary for its proper structure or biological function. As a result, the protein may not be able to perform its normal role.

38. Could a mutation affect protein function without greatly changing protein length?

- Yes. A mutation can replace only one amino acid while leaving the total length of the protein unchanged. However, that amino acid may be important for maintaining the protein's structure or interactions. In our experiment, the documented **BCKDHA c.929C>G (p.Thr310Arg)** mutation is an example because it changes one amino acid without changing the 445-amino-acid protein length.

39. Could a mutation cause disease without changing the protein sequence? Give a possible molecular mechanism.

- Yes. A mutation can affect processes such as gene regulation, RNA splicing, mRNA stability, or the amount of protein produced without changing the amino-acid sequence itself. For example, a regulatory mutation could decrease **BCKDHA** expression, resulting in less functional BCKD complex being available in the cell.

40. What evidence from your analysis supports the proposed molecular mechanism of your disease?

- Our sequence analysis showed that the documented **c.929C>G** substitution changes the **ACA** codon to **AGA**, producing the predicted **Thr310Arg (T310R)** substitution. The CDS remained **1,338 bp**, while the predicted protein remained **445 amino acids** long. Since there was no frameshift or premature stop codon, the analysis indicates that the main sequence-level effect is a single amino-acid substitution. Further experimental evidence is needed to determine its effect on BCKDHA function.

41. Which conclusions are supported directly by your computational results, and which require evidence from published experimental studies?

- The computational analysis directly demonstrates the nucleotide substitution, the **ACA→AGA** codon change, and the resulting predicted **Thr310Arg** amino-acid substitution. It also shows that the reading frame and protein length were maintained and that no premature stop codon was introduced. However, determining whether the mutation actually affects BCKDHA stability, assembly, enzyme activity, and MSUD requires findings from experimental and clinical studies.

XIV. GITHUB REPOSITORY AND GALAXY HISTORY

GITHUB REPOSITORY:

https://github.com/zenasdeguit/Deguit_MSUD_BCKDHA_Mutation_Lab

GALAXY HISTORY:

<https://usegalaxy.org/u/zenasdeguit/h/deguit-msud-bckdha-mutation-lab>

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